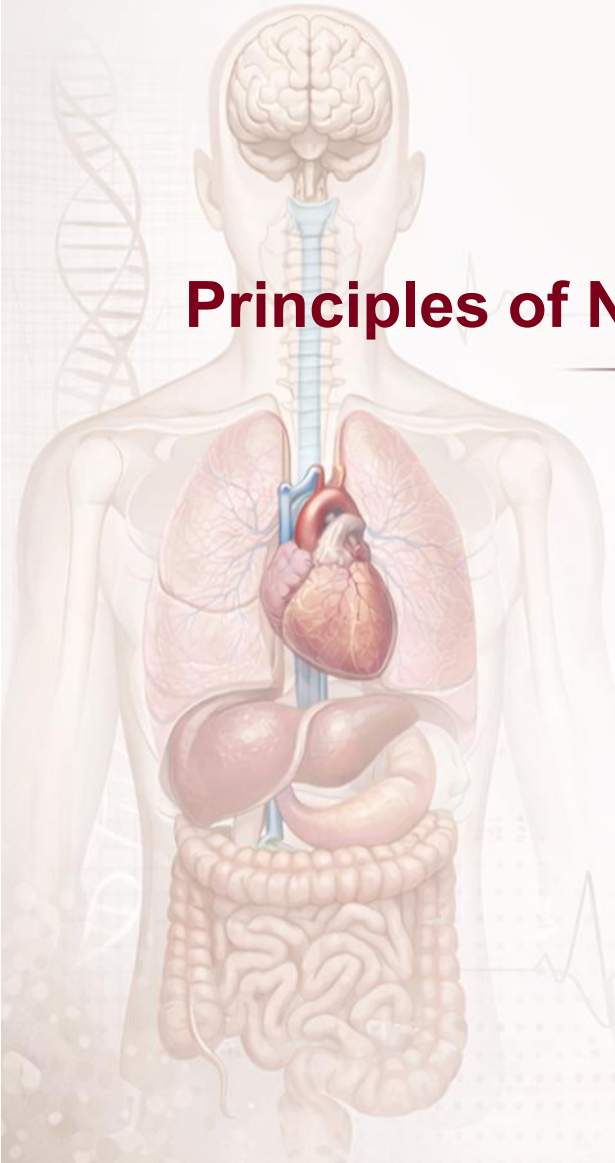
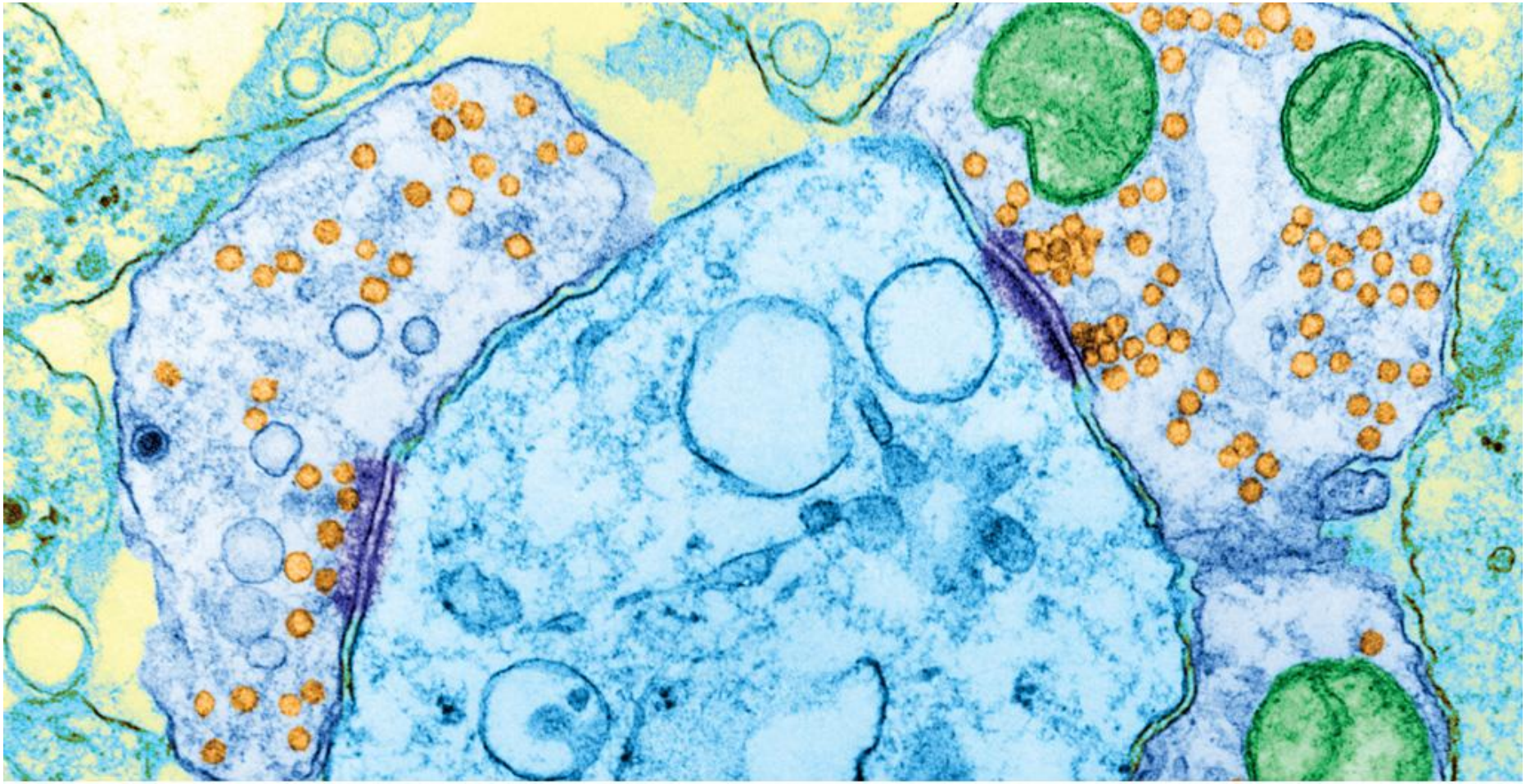


Chapter 4.
Principles of Neural and Hormonal Communication

KOREA UNIVERSITY
1905

Transmission electron micrograph of two synapses



Science Source

Nervous and endocrine systems

https://www.youtube.com/watch?v=dma-xY6WzN8

Neurons and hormones in terms of homeostasis

- To maintain homeostasis, cells must work in a coordinated fashion toward common goals
- The two major regulatory systems of the body that ensure life-sustaining coordinated responses are *nervous and endocrine systems*

- Neural communication is accomplished by nerve cells (neurons), specialized for rapid electrical signaling and secreting neurotransmitters, short-distance chemical messengers acting on nearby target organs (nervous system rapidly controls body's muscles and exocrine secretions)
- Hormonal communication is accomplished by hormones, long-distance chemical messengers secreted by the endocrine glands into the blood. The blood carries the hormones to distant target organs (endocrine system maintain the process that require duration, such as metabolic activities, water and electrolyte balance, and growth)

4.1 Introduction to Neural Communication

Introduction to neural communication

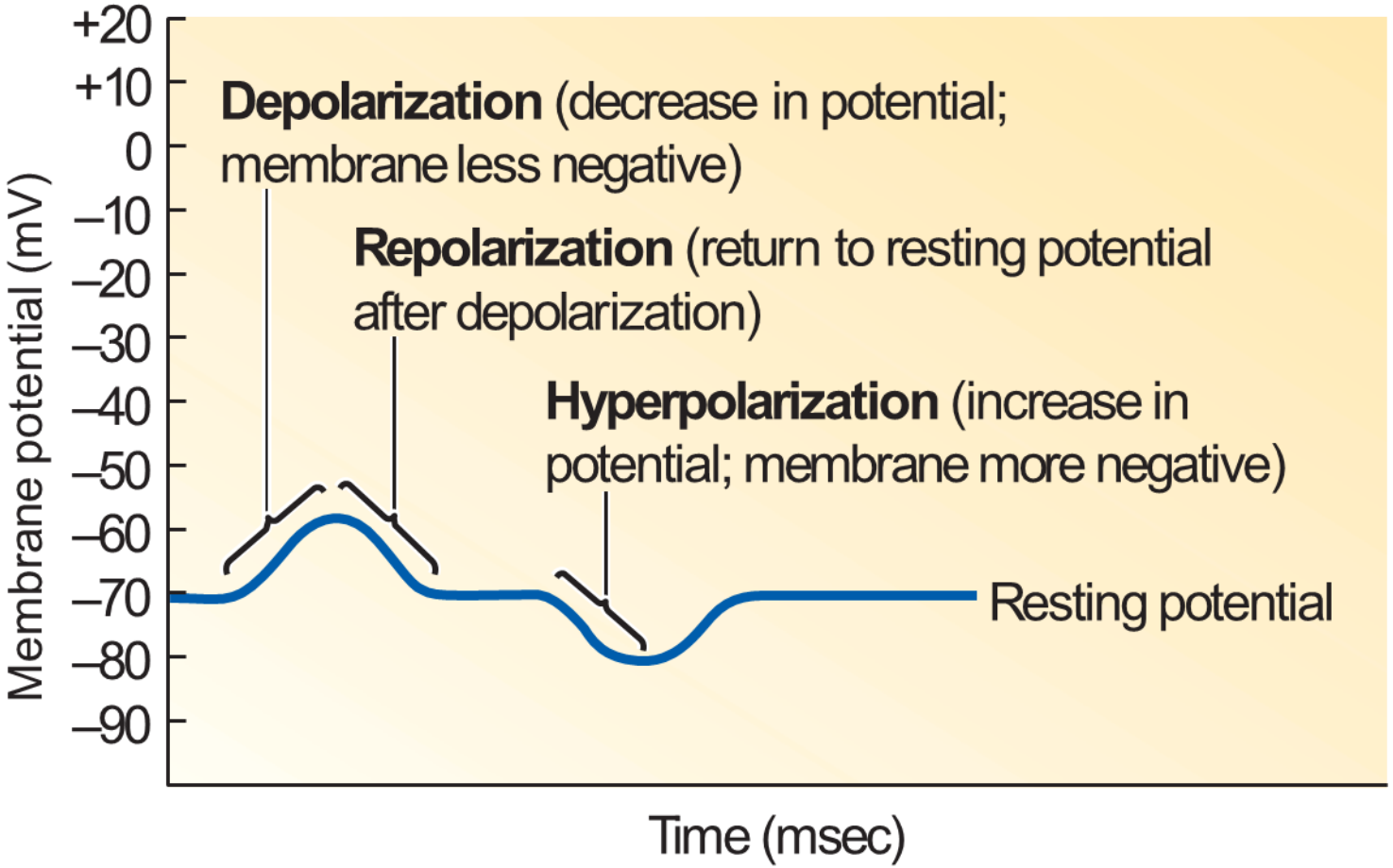
- As we learned from Chapter 3, all cells display a membrane potential, a separation of positive and negative charges across the membrane (uneven distribution of Na^+, K^+, and large intracellular protein anions in the ICF and ECF, and differential permeability of the plasma membrane to these ions)

- Unlike other cells at resting membrane potential, nerve and muscle are excitable tissues which undergoes a transient and rapid fluctuation in their membrane potential
- Nerve and muscle produce electrical signals when excited
  - Neurons use these to receive, process, initiate, and transmit messages
  - Muscle cells use these to initiate contraction

Introduction to neural communication

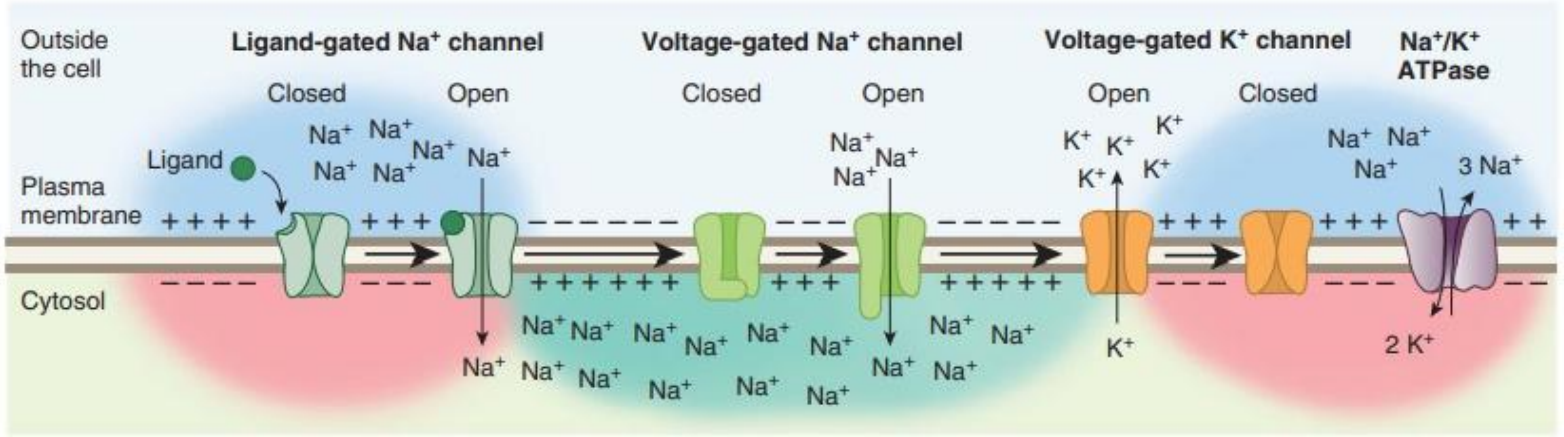
- Membrane potential becomes less negative during depolarization and more negative during hyperpolarization
- Key terms describing changes in potential
  - Polarization: membrane potential is not 0 mV (more polarized = extremely high or low potential)
  - Depolarization: potential becomes less polarized and less negative than resting potential
  - Repolarization: potential returns to resting potential after having been depolarized
  - Hyperpolarization: potential becomes more polarized and more negative than resting potential

Introduction to neural communication



Introduction to neural communication

- Membrane potential changes or electrical signals are produced by changes in ion movement across the plasma membrane
  - Net inward flow of cations causes membrane depolarization (less negative inside)
  - Net outward low of cations causes hyperpolarization



https://www.pharmacy180.com/article/cell-membrane-potential-3544/

Introduction to neural communication

- A triggering event causes a change in membrane potential
  - It changes membrane potential by altering the membrane permeability and consequently altering ion flow across the membrane
  - Water-soluble ions, responsible for charge redistribution, cannot penetrate lipid bilayer -> channel- or carrier-mediated transport
  - Four types of gated channels: voltage-gated (membrane potential), chemically gated (chemical messenger), mechanically gated (mechanical force), and thermally gated (temperature)
  - Triggering event -> re-shaping of gated channel

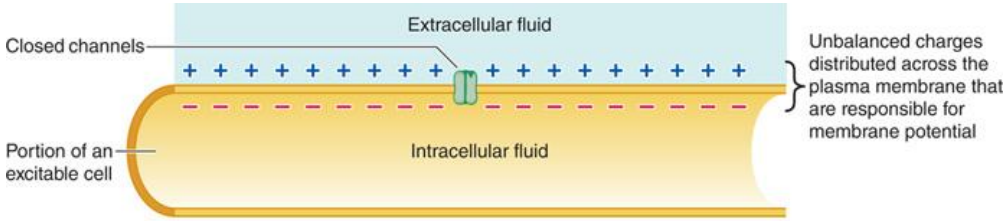
4.2 Graded Potentials

Two forms of electrical signals

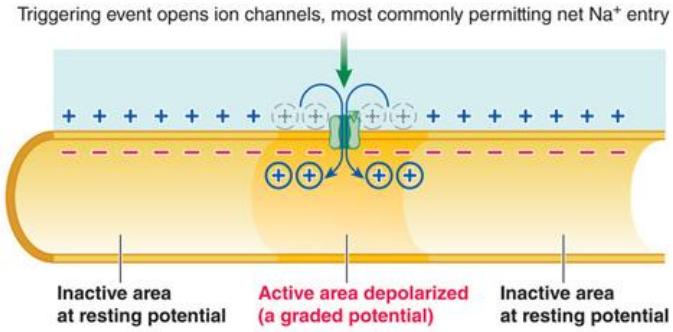
- Graded potentials serve as short-distance signal, while action potential signal over long distances

- Graded potentials are local changes in membrane potential
- Occur in varying grades or degrees of magnitude or strength
  - The stronger a triggering event, the larger the resultant graded potential
  - The longer a triggering event, the longer the duration of potential
- Spread by passive current flow
  - Current: any flow of electrical charges
  - Resistance: hindrance to electrical charge movement
- Graded potential die out over short distances

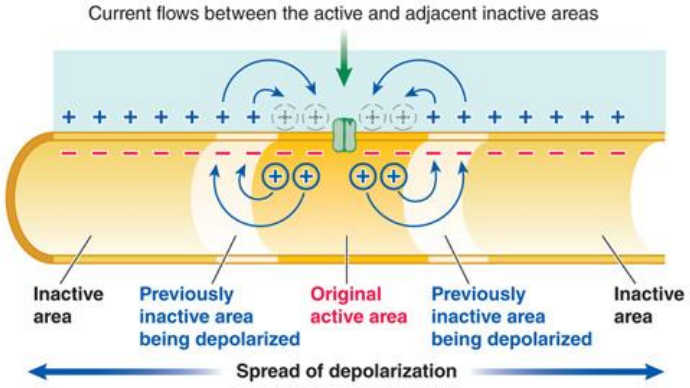
Graded potentials



(a) Entire membrane at resting potential



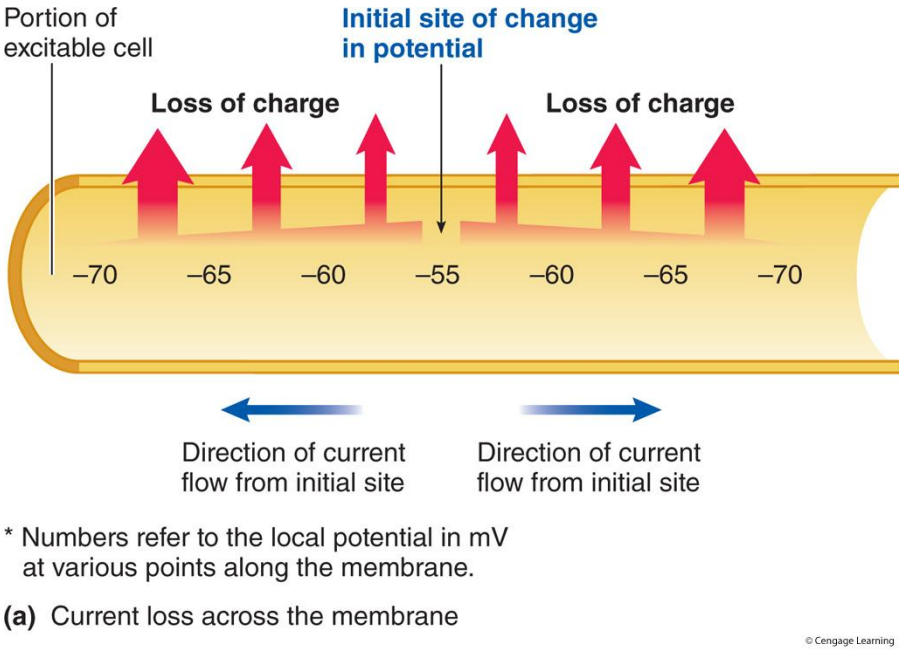
(b) Inward movement of Na^+ depolarizes membrane, producing a graded potential

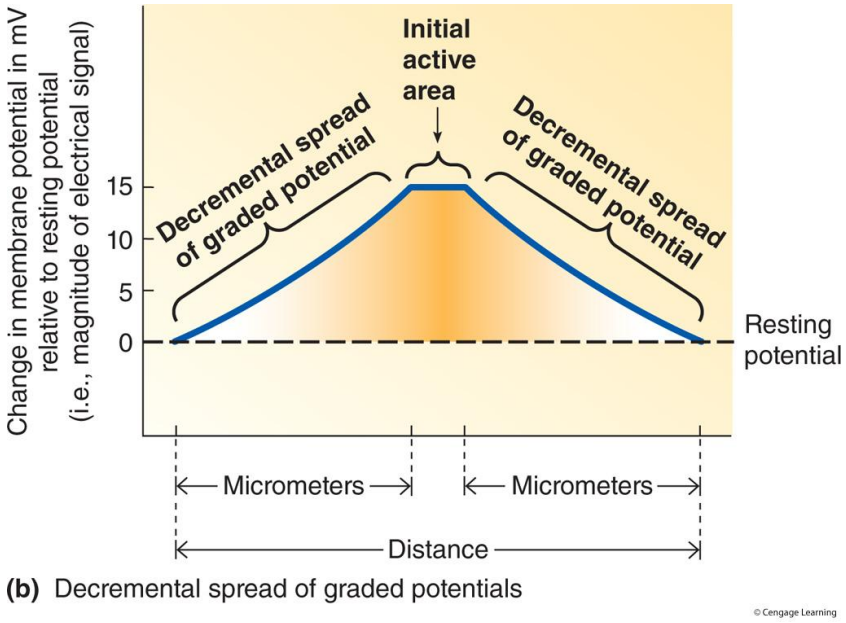


(c) Depolarization spreads by local current flow to adjacent inactive areas, away from point of origin

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Graded potentials



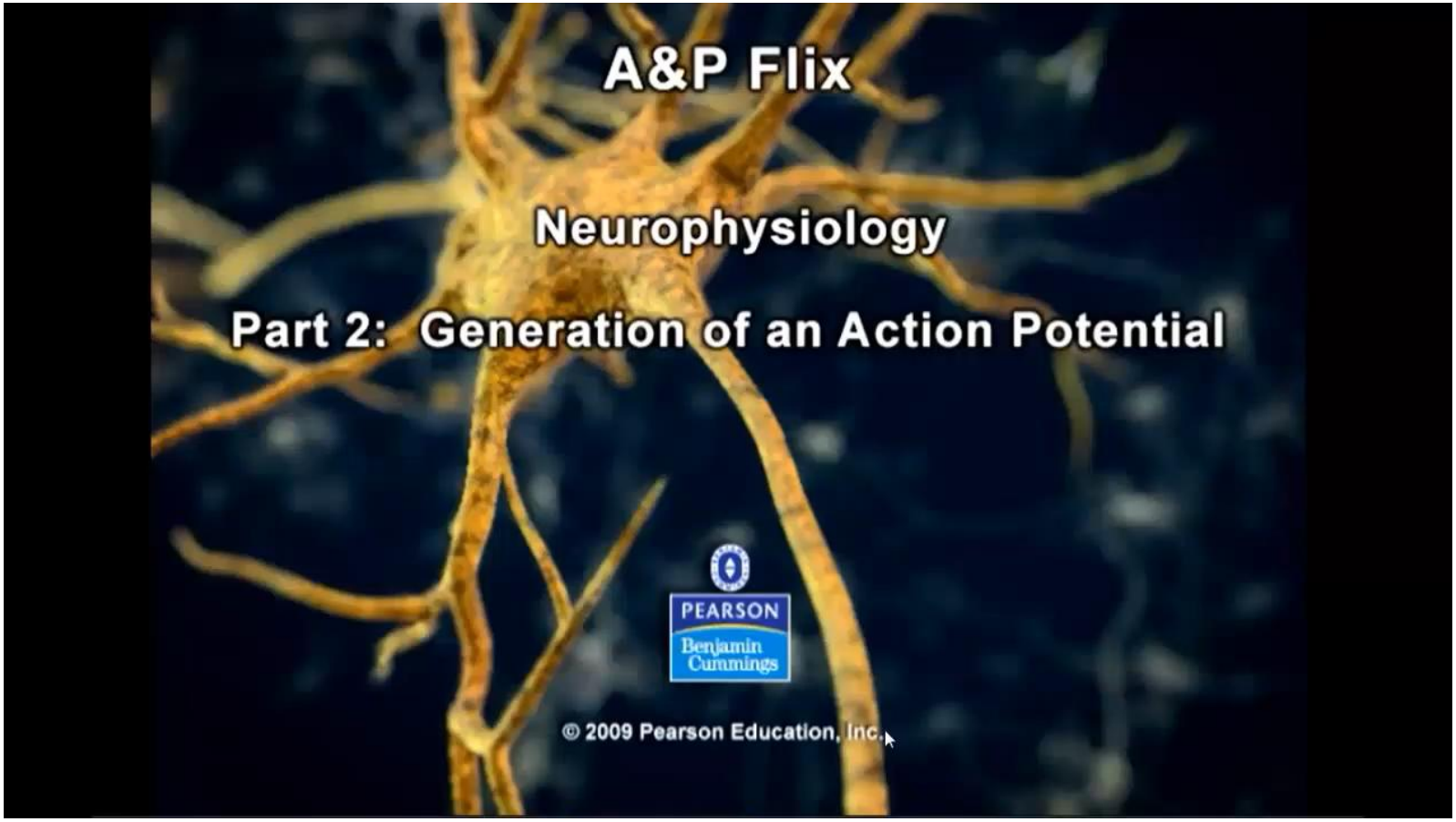


4.3 Action Potentials

Action potentials

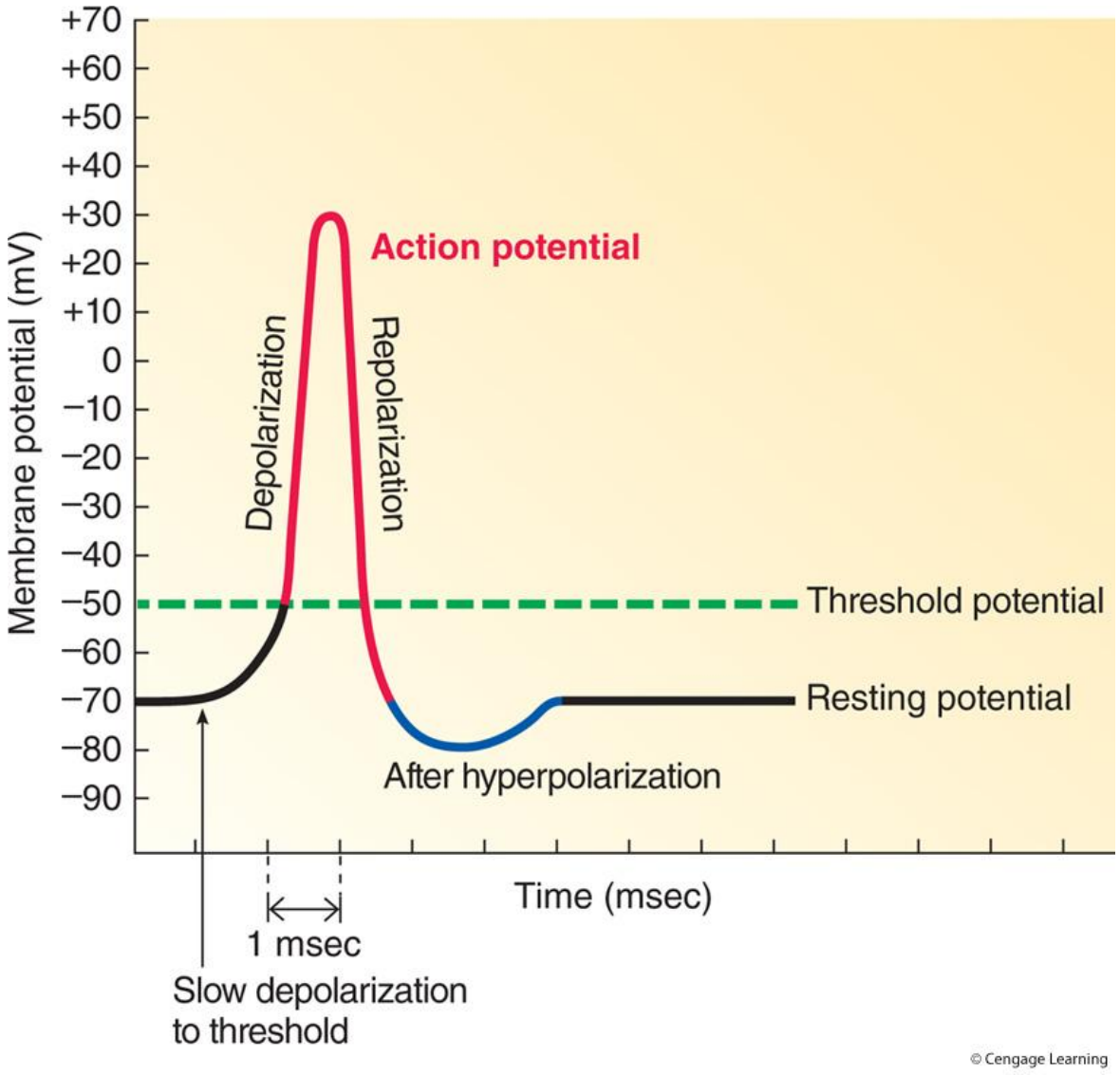
- Compared to graded potentials…
  - Action potentials are brief, rapid, large (100 mV) changes in membrane potential
  - Action potential actually reverses the charge
    - Inside of the excitable cell (neuron, muscle cell) transiently becomes more positive than the outside
  - Changes in membrane permeability and ion movement lead to an action potential
    - Voltage-gated Na^+ and K^+ channels
    - Changes in permeability and ion movement during an action potential

Action potentials

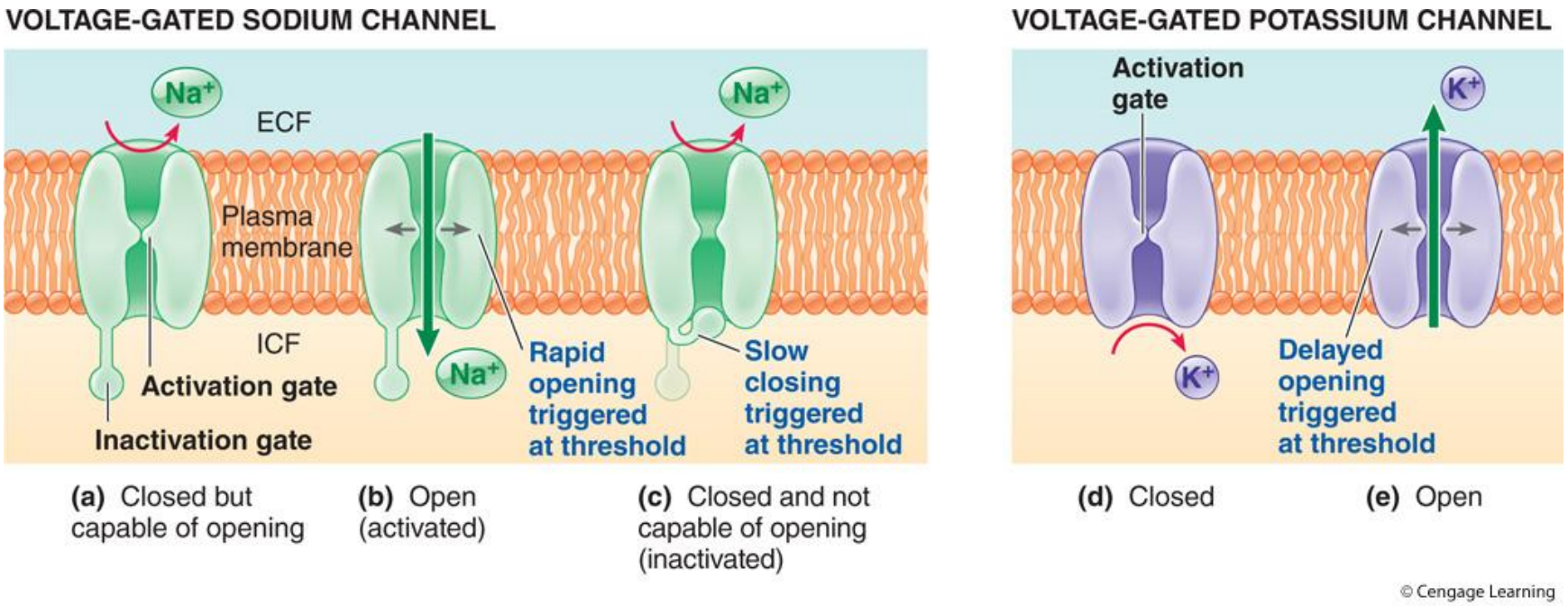


https://www.youtube.com/watch?v=-h_kWFM2faQ

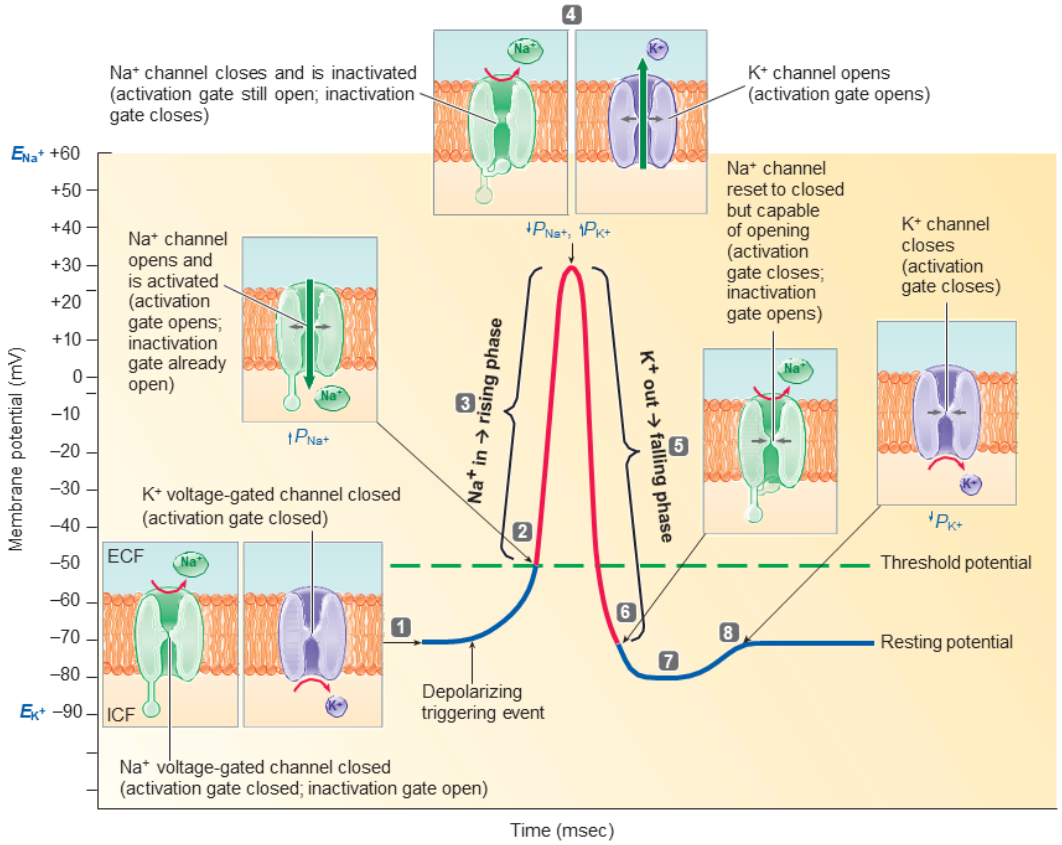
Action potentials



Voltage-gated channels in action potential



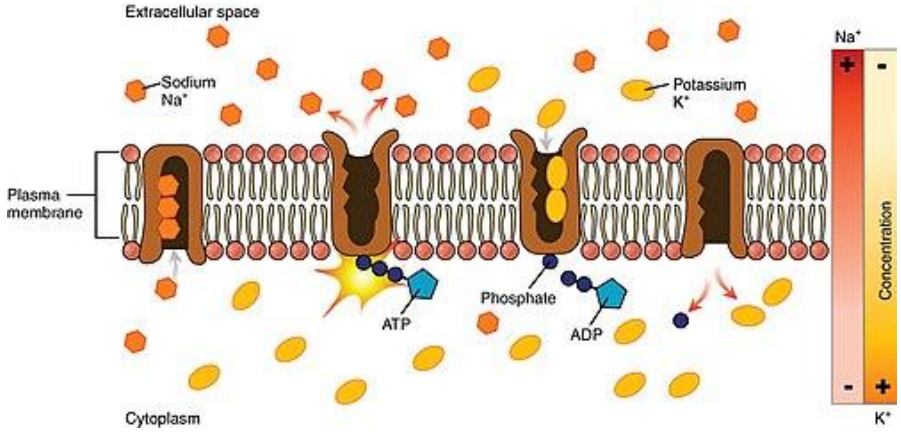
Permeability changes and ion fluxes in action potential



1. Resting potential: all voltage-gated channels closed.
2. At threshold, Na+ activation gate opens and PNa+ rises.
3. Na+ enters cell, causing explosive depolarization to +30 mV, which generates rising phase of action potential.
4. At peak of action potential, Na+ inactivation gate closes and PNa+ falls, ending net movement of Na+ into cell. At the same time, K+ activation gate opens and PK+ rises.
5. K+ leaves cell, causing its repolarization to resting potential, which generates falling phase of action potential.
6. On return to resting potential, Na+ activation gate closes and inactivation gate opens, resetting channel to respond to another depolarizing triggering event.
7. Further outward movement of K+ through still-open K+ channel briefly hyperpolarizes membrane, which generates after hyperpolarization.
8. K+ activation gate closes, and membrane returns to resting potential.

Action potentials

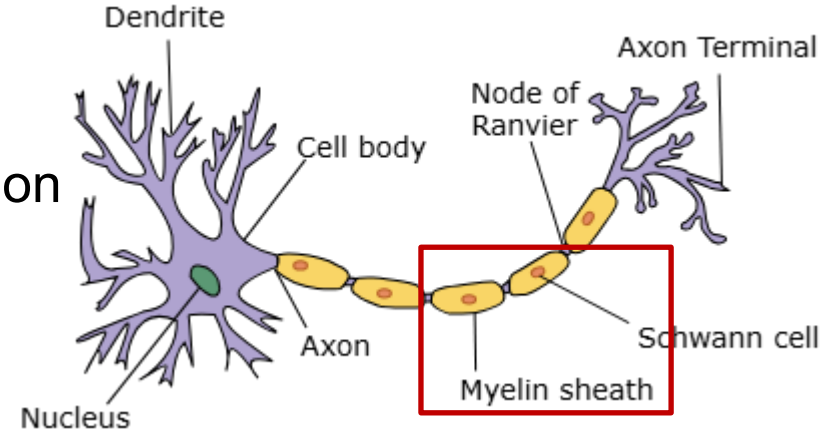
- Na^+–K^+ pump gradually restores concentration gradients disrupted by action potentials
  - At the completion of an action potential, membrane potential restores to resting
    - Ion distribution has been altered slightly

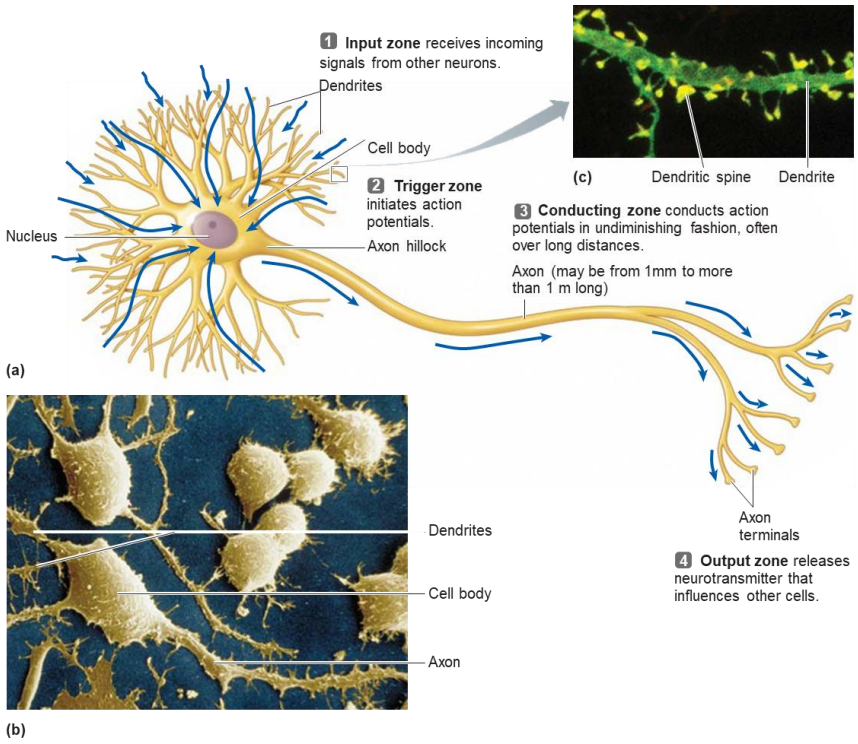


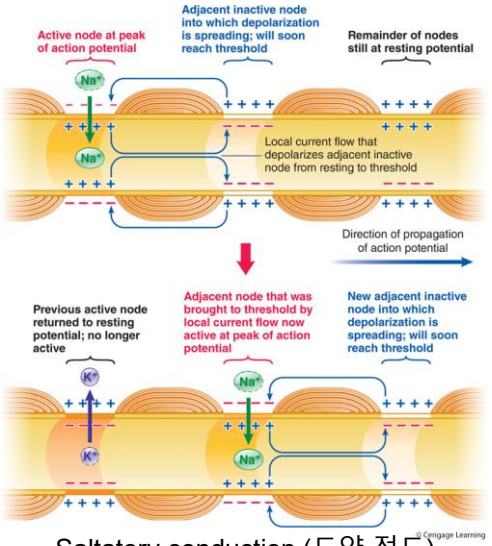
Sodium-potassium ATPase pump

Action potentials

- Action potentials are propagated from the axon hillock to the axon terminals
  - Release chemical messengers
  - Myelination and saltatory conduction



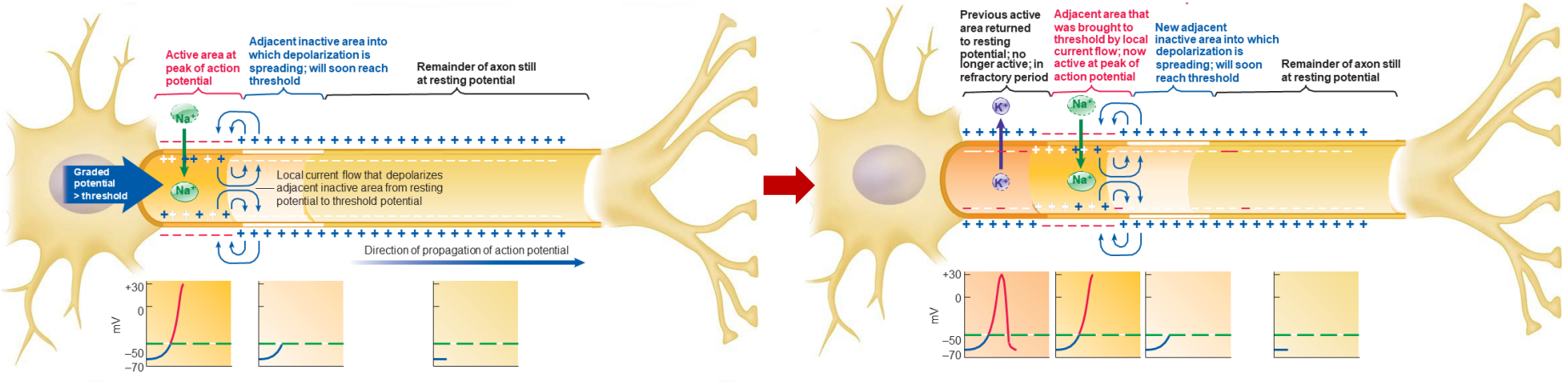




Saltatory conduction (도약 전도)
rapid, "jumping" propagation of action potentials along myelinated axons

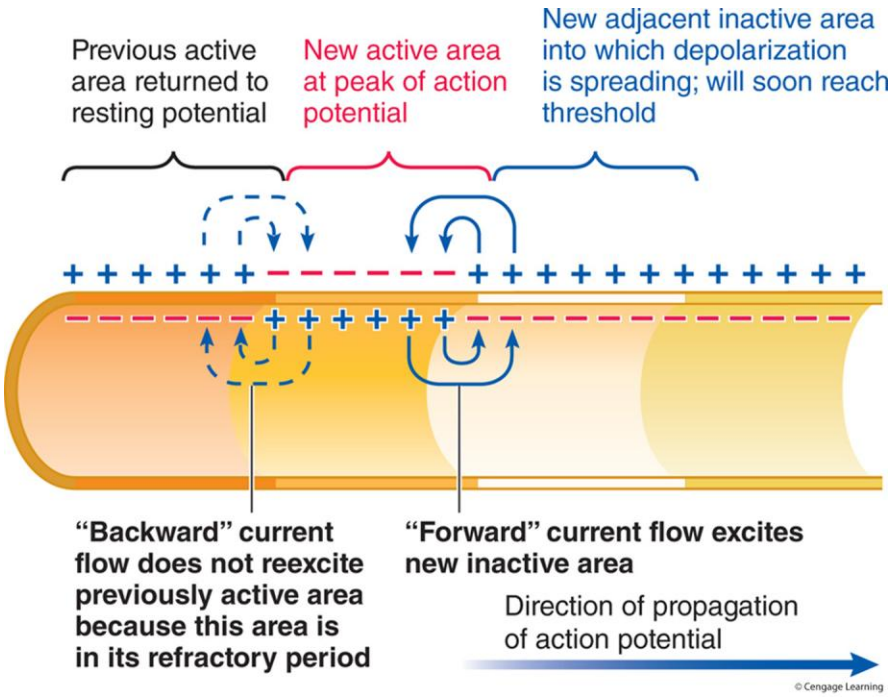
Contiguous conduction of action potential

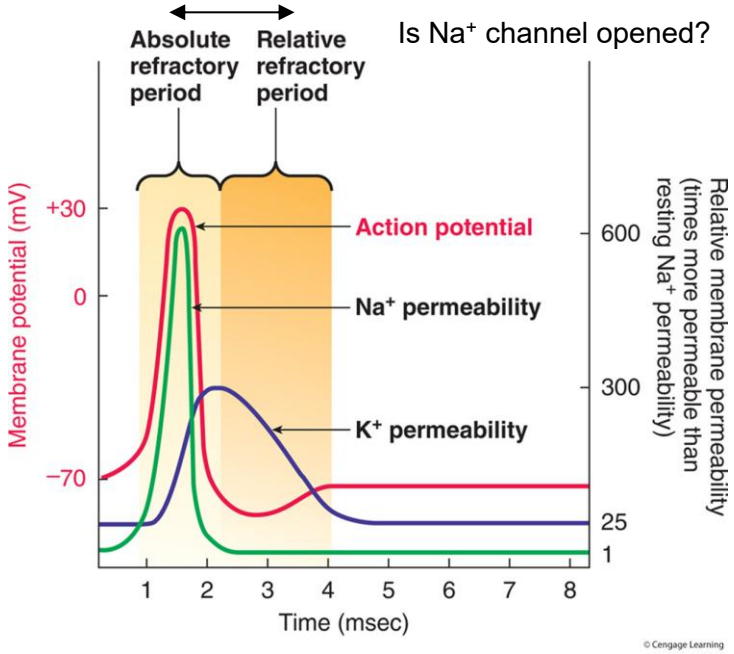
- Once initiated, action potentials are conducted throughout a nerve fiber
  - Contiguous conduction (연속 전도)



Value of refractory period in action potential

- Refractory period ensures one-way propagation of action potentials and limits their frequency
  - Action potential cannot be initiated in a region that has just undergone an action potential
    - Absolute and relative refractory periods





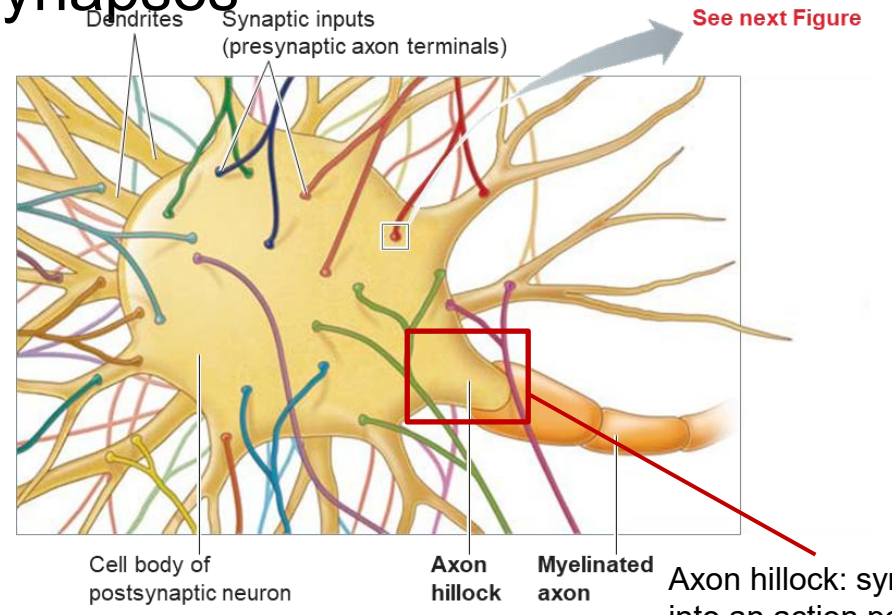
Action potential characteristics

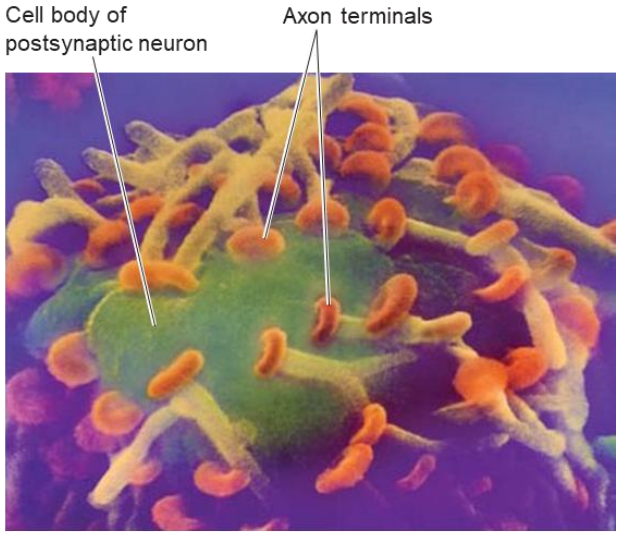
- Occur in all-or-none fashion
- Strength of a stimulus is coded by the frequency of action potentials
- Myelination increases the speed of conduction of action potentials
- Fiber diameter influences the velocity of action potential propagation

4.4 Synapses and Neuronal Integration

Definition and types of synapses

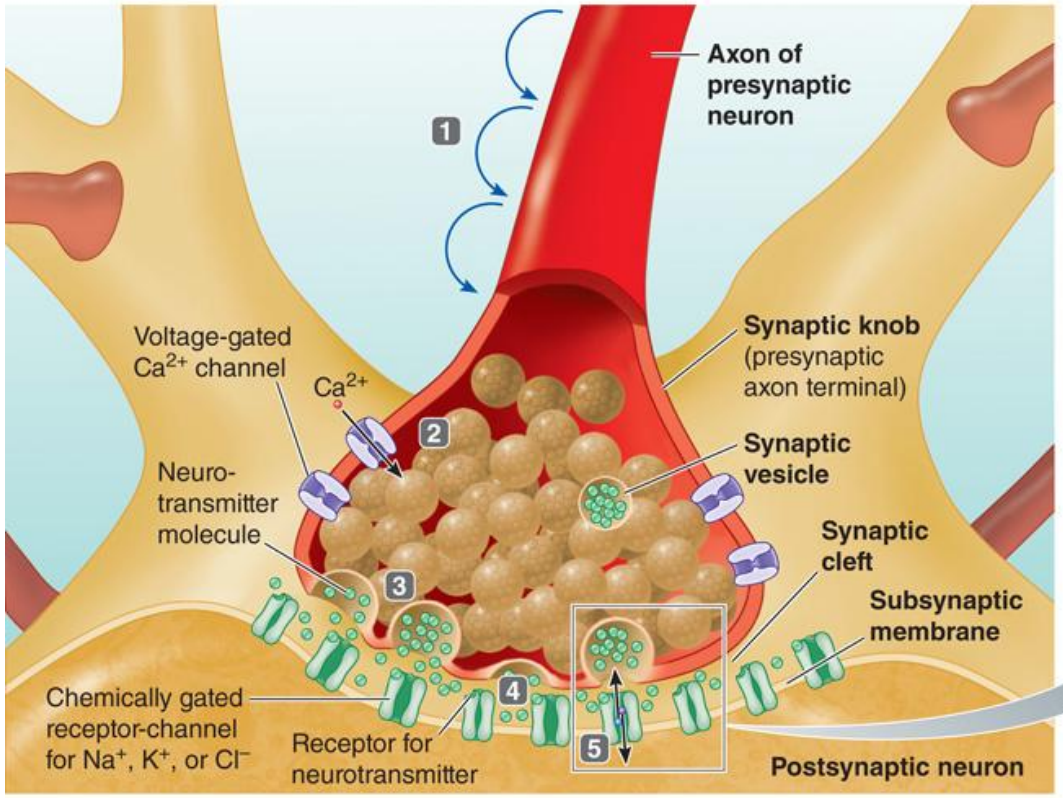
- Synapse: junction between neurons (pre- and post-synaptic neurons)
  - Electrical synapses: neurons connected directly by gap junctions
  - Chemical synapses: chemical messenger transmits information one way across a space separating the two neurons
- Most synapses in the human nervous system are chemical synapses





Axon hillock: synaptic information is integrated and converted into an action potential -> decides whether to output a signal

Structure and function of a single synapse



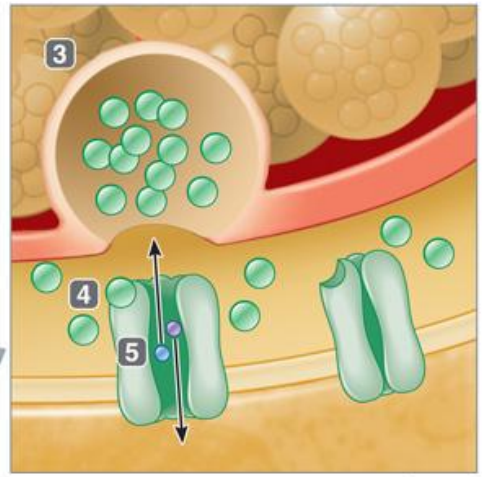
1 Action potential reaches axon terminal of presynaptic neuron.

2 Ca^{2+} enters synaptic knob (presynaptic axon terminal).

3 Neurotransmitter is released by exocytosis into synaptic cleft.

4 Neurotransmitter binds to receptors that are an integral part of chemically gated channels on subsynaptic membrane of postsynaptic neuron.

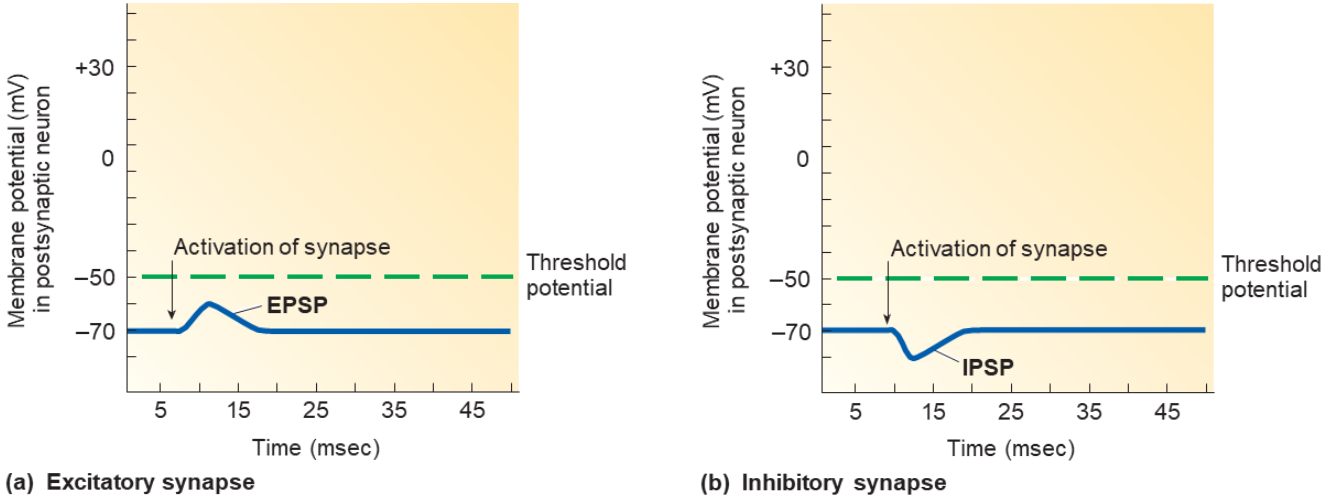
5 Binding of neurotransmitter to receptor-channel opens that specific channel.



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Synapses and neuronal integration

- Neurotransmitter carries the signal across a synapse
  - Receptor channels: combined receptor plus channel units
- Some synapses excite, whereas others inhibit, the postsynaptic neuron
  - Excitatory synapses -> EPSP (excitatory postsynaptic potential)
  - Inhibitory synapses -> IPSP (inhibitory postsynaptic potential)

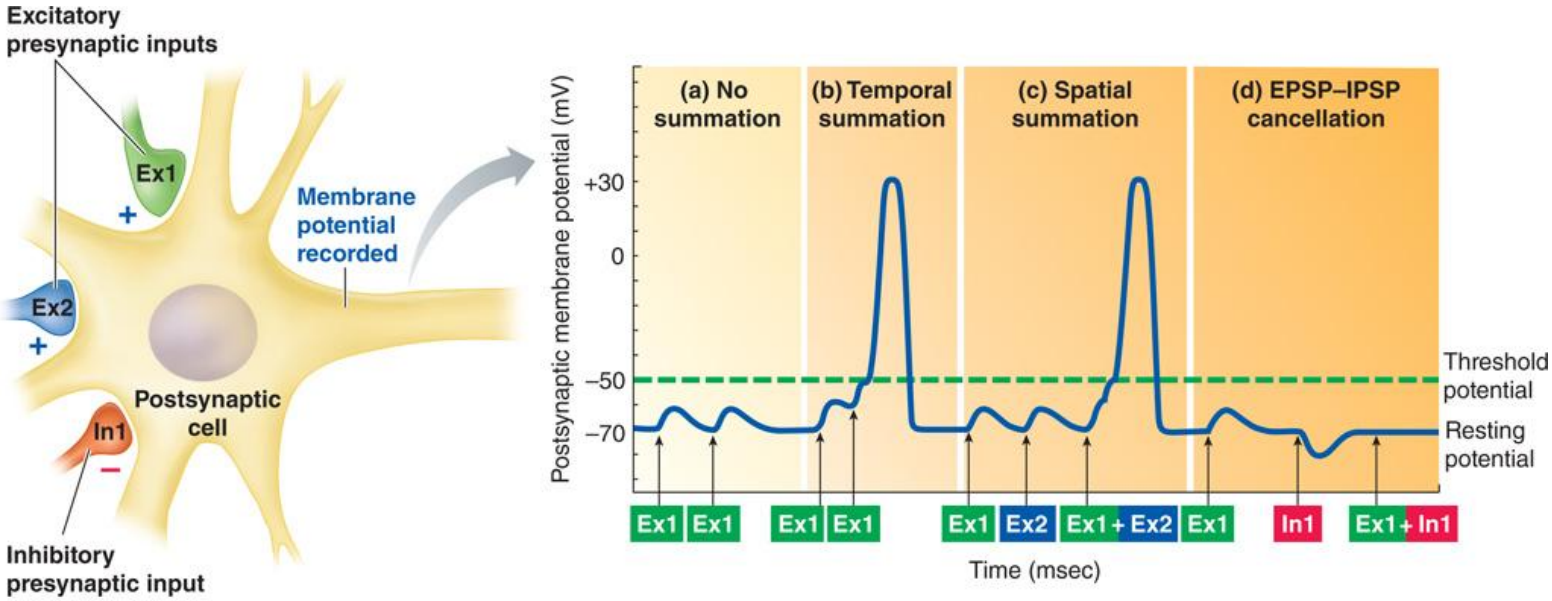


- Each neurotransmitter–receptor combination always produces the same response

Synapses and neuronal integration

- Neurotransmitters are quickly removed from the synaptic cleft
  - After combining with the postsynaptic receptor-channel, chemical transmitters are removed and the response is terminated
- Grand postsynaptic potential depends on the sum of all presynaptic input activities
  - Temporal and spatial summation
  - Cancellation of concurrent EPSPs and IPSPs
- Importance of postsynaptic integration
  - Information processing
  - Decision making
  - Precision of neural circuits

Determination of grand postsynaptic potential



(a) If an excitatory presynaptic input (Ex1) is stimulated a second time after the first EPSP in the postsynaptic cell has died off, a second EPSP of the same magnitude will occur.

(b) If, however, Ex1 is stimulated a second time before the first EPSP has died off, the second EPSP will add onto, or sum with, the first EPSP, resulting in *temporal summation*, which may bring the postsynaptic cell to threshold.

(c) The postsynaptic cell may also be brought to threshold by *spatial summation* of EPSPs that are initiated by simultaneous activation of two (Ex1 and Ex2) or more excitatory presynaptic inputs.

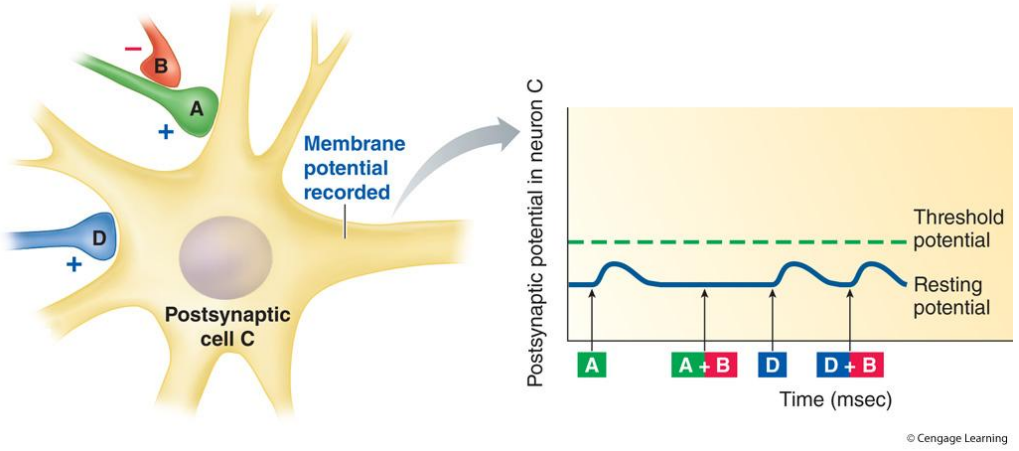
(d) Simultaneous activation of an excitatory (Ex1) and inhibitory (In1) presynaptic input does not change the postsynaptic potential, because the resultant EPSP and IPSP cancel each other out.

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Modulators other than neurotransmitters

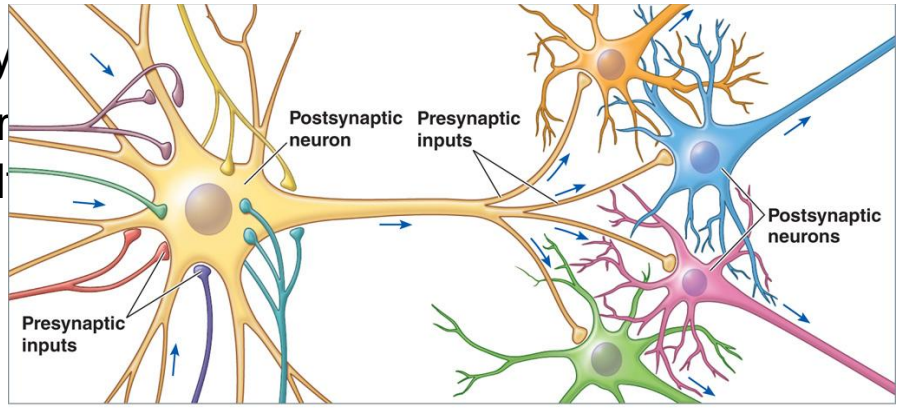
- Neuromodulators
  - Chemical messengers that do not cause the formation of EPSPs or IPSPs but instead act slowly to bring about long-term changes that subtly modulate (depress or enhance) the action of the synapse

- Presynaptic modulation
  - Presynaptic inhibition: sometimes, a third neuron activity influences synaptic activity to decrease the amount of neurotransmitter released by presynaptic axon terminal
  - Presynaptic facilitation: otherwise, a third neuron activity influences synaptic activity to enhance the amount of neurotransmitter released by presynaptic axon terminal



Synapses and neuronal integration

- Drugs and diseases can modify synaptic transmission
  - Most drugs that influence the nervous system function by altering synaptic mechanisms
    - Alter the synthesis, storage, or release of neurotransmitters
    - Modify neurotransmitter interaction with postsynaptic receptor
    - Influence neurotransmitter reuptake or destruction
    - Replace deficient neurotransmitter with substitute drug
- Neurons are linked through complex converging and diverging pathways
  - Convergence: a neuron has many
  - Divergence: branching of axon ter
    synapses with and influences mul

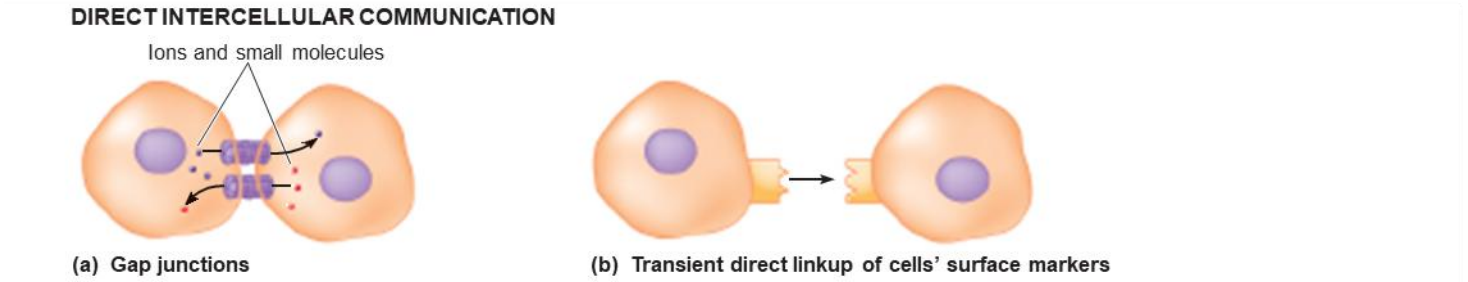


4.5 Intercellular Communication and Signal Transduction

Intercellular communication and signal transduction

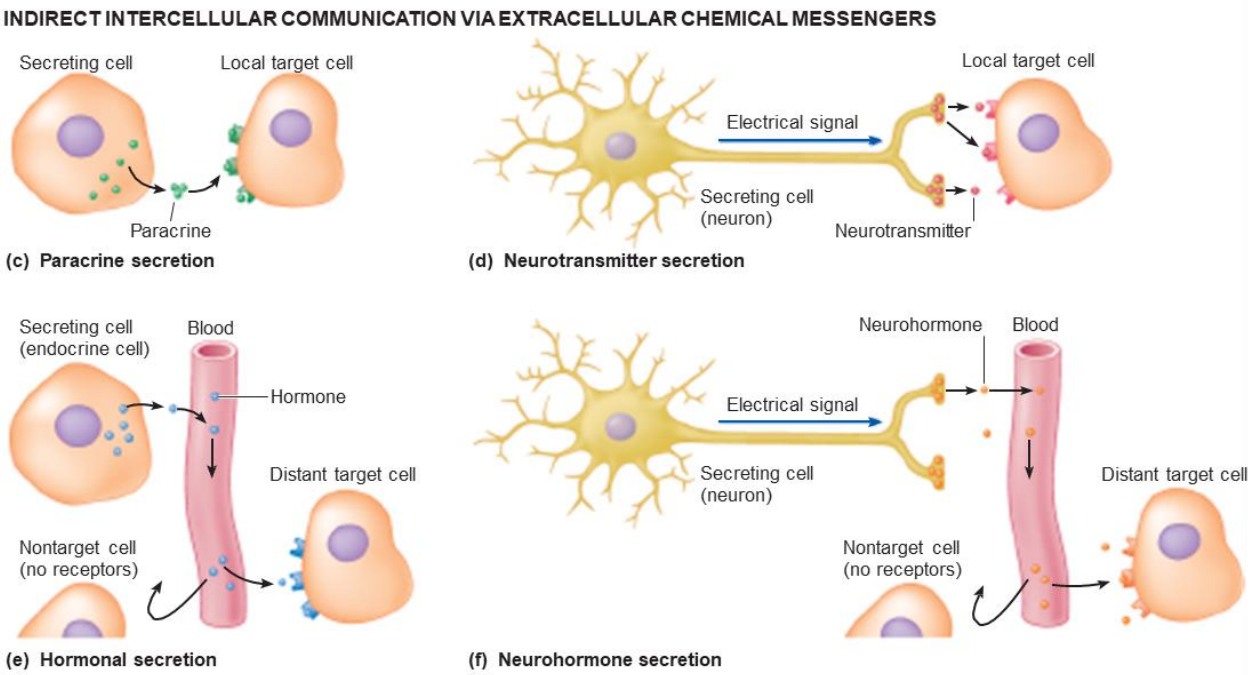
- Coordination of the diverse activities of cells in our body to accomplish life-sustaining responses depends on the ability of cells to communicate with another

- Direct intercellular communication is accomplished through gap junctions, possibly through tunneling nanotubes, and through linkup of surface markers



Intercellular communication and signal transduction

- Communication among cells is largely orchestrated by extracellular chemical messengers
  - Most common means by which cells communicate with one another: indirectly through extracellular chemical messengers, or signal molecules



Extracellular chemical messengers (signal molecules)

- Neurotransmitters: very short-range chemical messengers released by neurons
- Paracrine/autocrines: local chemical messengers
- Hormones: long-range chemical messengers secreted into the blood by endocrine glands
- Neurohormones: hormones released into blood by neurosecretory neurons

- All of them are synthesized by specialized controlling cells and released into ECF by appropriate stimulation to act on the messenger's target cells
  - Incoming signals are transmitted to the target cell, where they are transformed into a specific cellular response

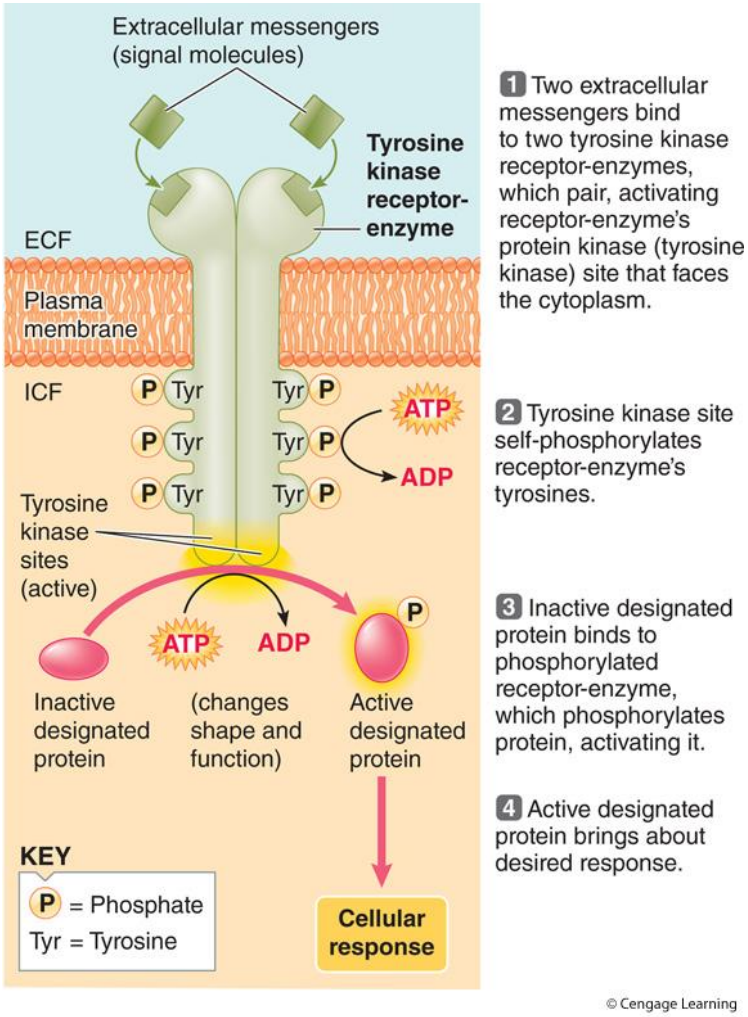
Signal transduction

- Signal transduction: the process by which incoming signals (instructed by extracellular chemical messengers) are transmitted to the target cell, where they are transformed into the specific cellular response
- Occur by different mechanisms
  - Lipid-soluble extracellular chemical messengers: cholesterol-derived steroid hormones, can enter plasma membrane to regulate gene expression
  - Water-soluble extracellular chemical messengers: peptide hormones, neurotransmitters, and local paracrines, have to first bind to receptor to trigger downstream cellular effects

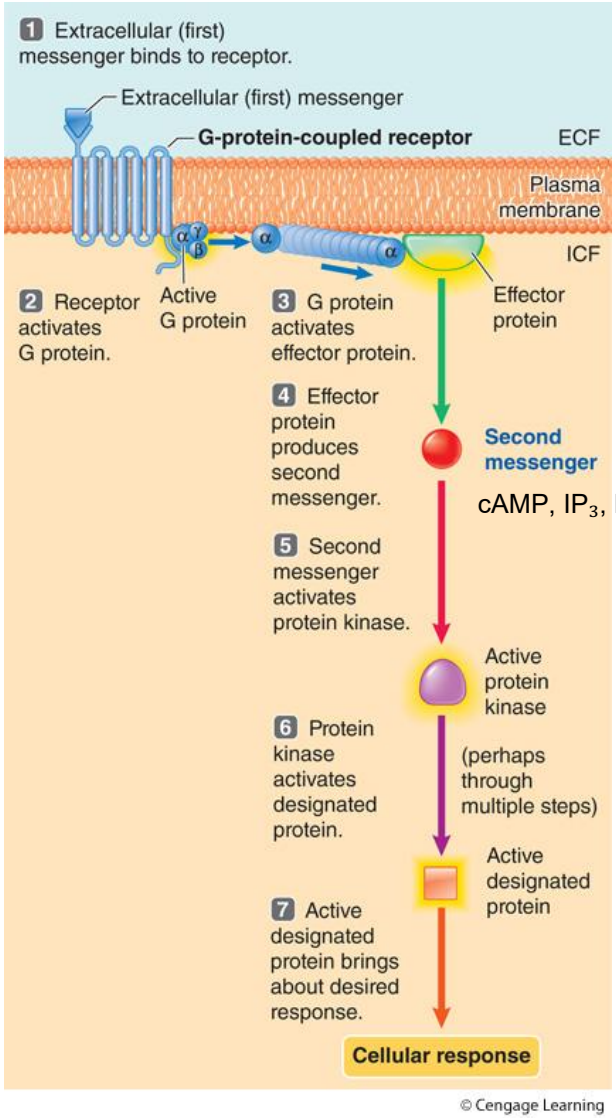
Water-soluble extracellular messengers during signal transduction

- 1) Some messengers bind to chemically gated receptor-channel to open or close the channel
- 2) Some messengers activate receptor-enzymes
  - Tyrosine kinase pathway, phosphorylates a set of proteins for cellular response
  - JAK/STAT pathway
- 3) Most water-soluble extracellular chemical messengers activate second-messenger pathways via G-protein-coupled receptors

Water-soluble extracellular messengers during signal transduction



Tyrosine kinase pathway

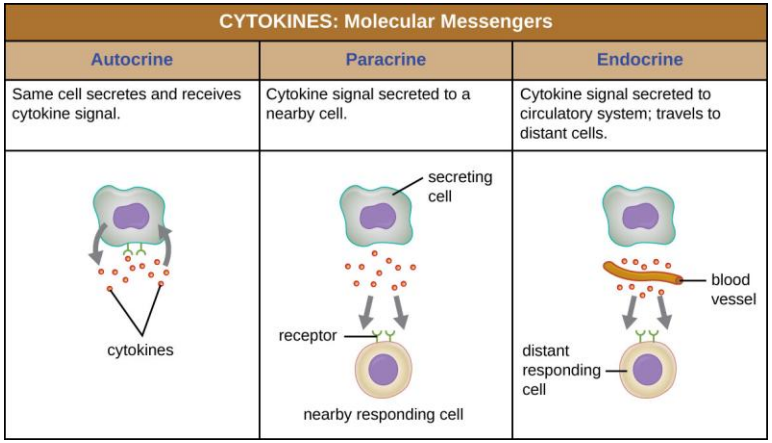


GPCR pathway

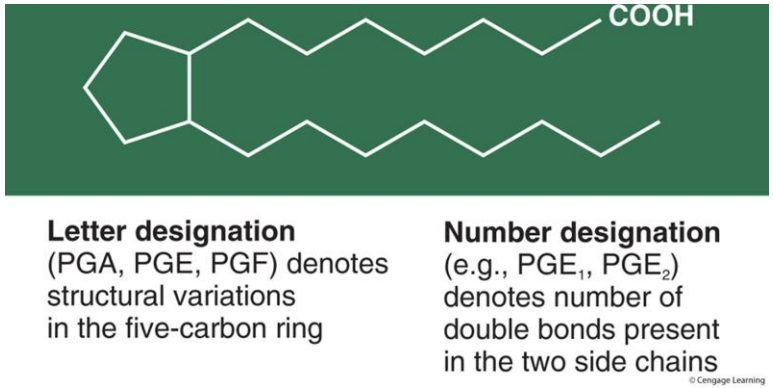
4.6 Introduction to Paracrine Communication

Paracrine communication

- Most paracrines are cytokines or eicosanoids
- Cytokines: collection of protein signal molecules that largely act locally to regulate immune response
- Eicosanoids: lipid signal molecules derived from fatty acid in the plasma membrane that act locally to regulate diverse cellular processes in the body



Diverse cytokines



Eicosanoid example: prostaglandin

4.7 Introduction to Hormonal Communication

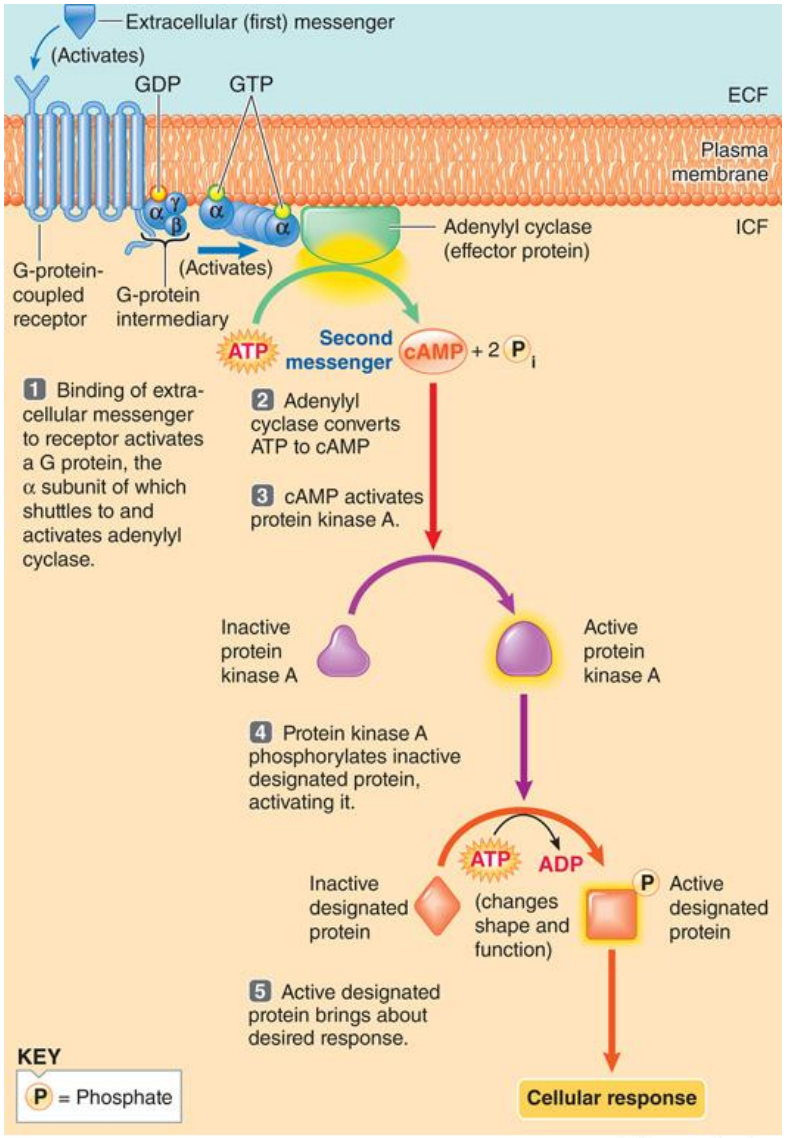
Introduction to hormonal communication

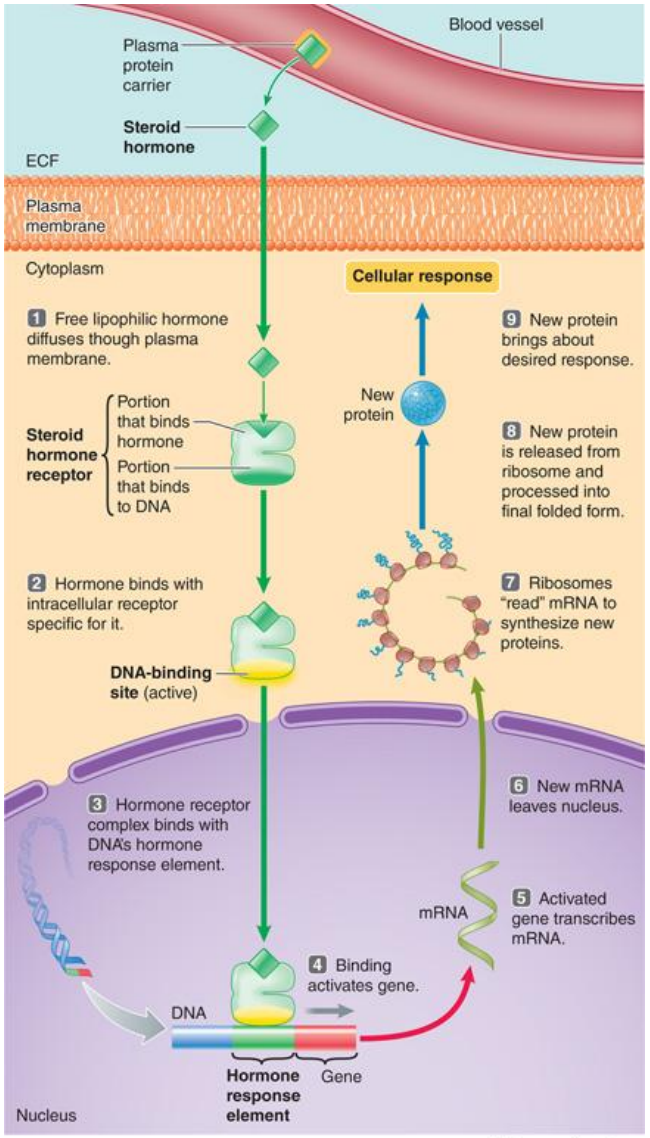
- Endocrinology: study of homeostatic chemical adjustments and other activities accomplished by hormones
  - Hormones are secreted by the endocrine glands into the blood
  - Two types based on solubility and chemical structure: hydrophilic or lipophilic
    - Hydrophilic hormones dissolve in the plasma (peptides)
    - Lipophilic hormones are transported bound to plasma proteins (steroids)
  - Mechanisms of synthesis, storage, and secretion of hormones vary according to chemical differences
  - Hormones generally produce their effect by altering intracellular proteins
    - Location of receptors for hydrophilic and lipophilic hormones
    - General means of hydrophilic and lipophilic hormone action

Introduction to hormonal communication

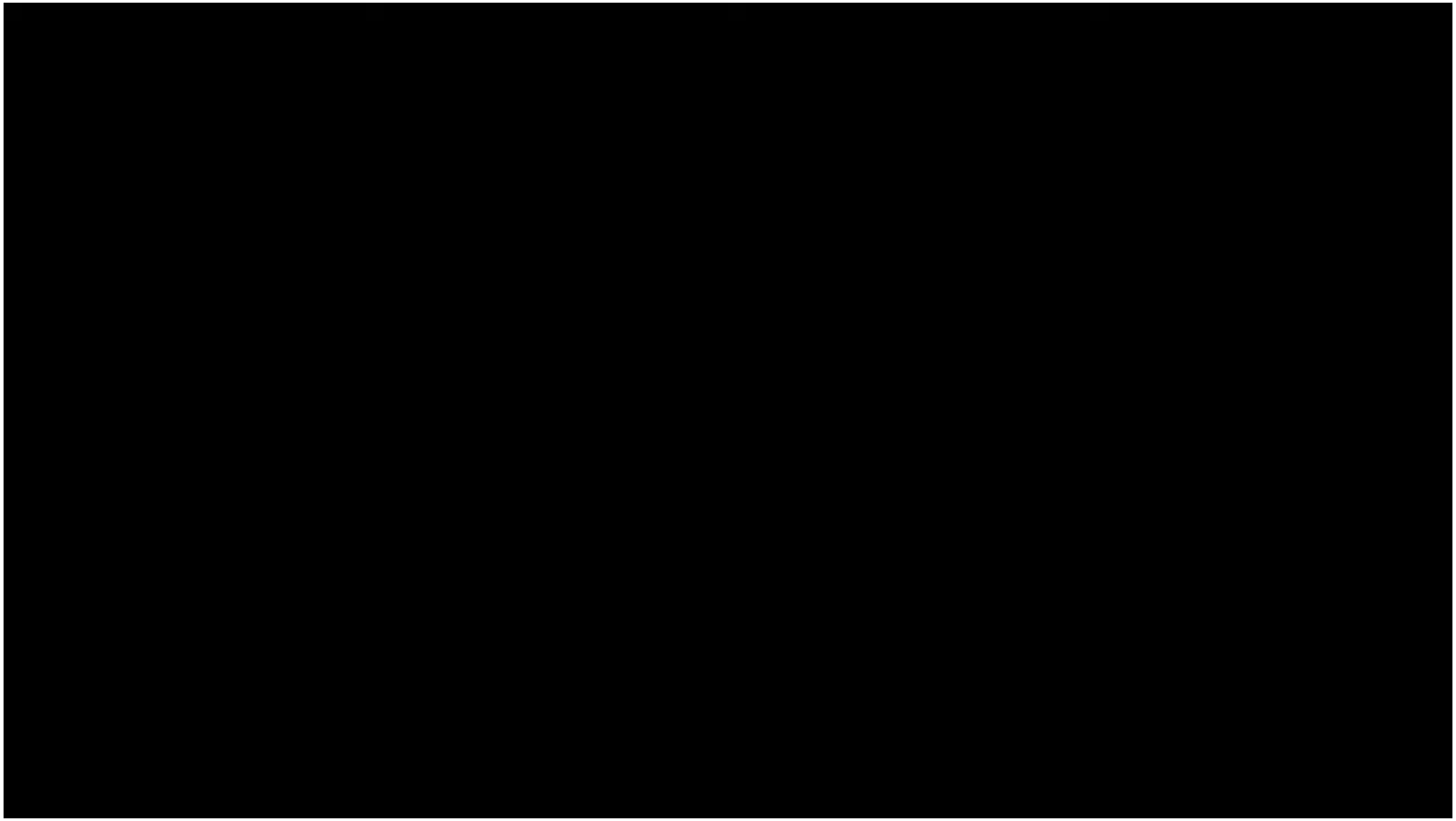
- Hydrophilic hormones alter preexisting proteins via second-messenger systems
  - Cyclic AMP second-messenger pathway
  - Ca^{2+} second-messenger pathway
  - Amplification by a second-messenger pathway
  - Regulation of receptors
- By stimulating genes, lipophilic hormones promote synthesis of new proteins

Hydrophilic and lipophilic hormones

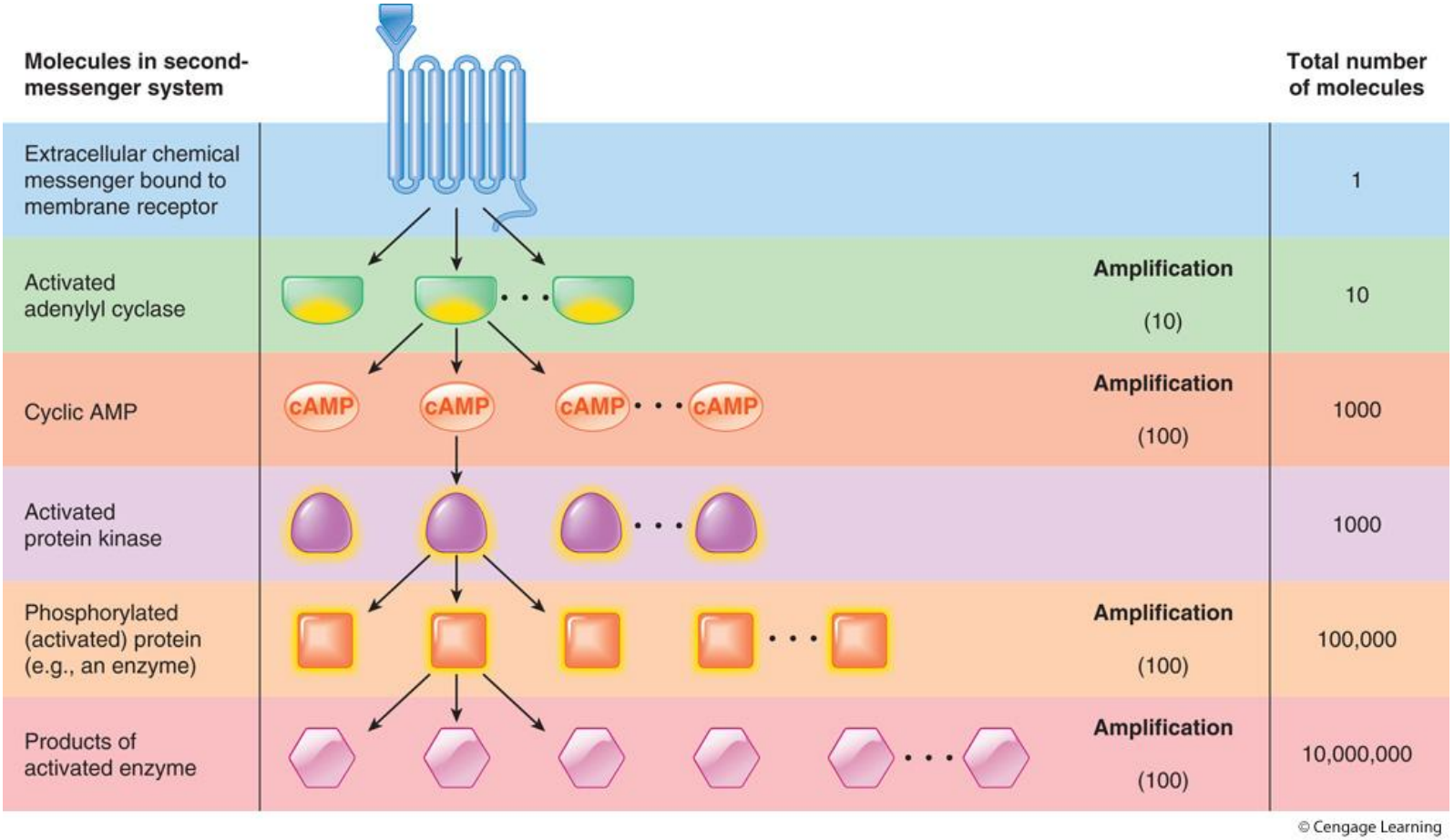




Specific example – cAMP-GPCR pathway



Second-messenger systems



Hydrophilic hormones alter preexisting proteins via second-messenger systems cAMP, IP_3, DAG, Ca^{2+}

Second-messenger systems allow signal amplification, rapid response, and diversification of cellular effects through intracellular signaling cascades

4.8 Comparison of the Nervous and Endocrine Systems

Comparison of the nervous and endocrine systems

- Nervous and endocrine systems are both main regulatory systems of the body
- Nervous system transmits electrical signals to the skeletal muscles and exocrine glands that it innervates, via synaptic transmission
  - Wired: transmission of signals via anatomically linked structure
  - Neural specificity is a result of anatomic proximity
- Endocrine system secretes hormones into the blood for delivery to distant sites of action
  - Wireless: endocrine glands are not anatomically linked with target cells
  - Endocrine specificity is a result of receptor specialization

Comparison of the nervous and endocrine systems

TABLE 4-6 Comparison of the Nervous System and the Endocrine System

Property	Nervous System	Endocrine System
Anatomic arrangement	A "wired" system: A specific structural arrangement exists between neurons and their target cells, with structural continuity in the system	A "wireless" system: Endocrine glands are widely dispersed and not structurally related to one another or to their target cells
Type of chemical messenger	Neurotransmitters released into the synaptic cleft	Hormones released into the blood
Distance of action of the chemical messenger	Short distance (diffuses across the synaptic cleft)	Long distance (carried by the blood)
Specificity of action on the target cell	Dependent on the close anatomic relationship between neurons and their target cells	Dependent on the specificity of target cell binding and responsiveness to a particular hormone
Speed of response	Generally rapid (milliseconds)	Generally slow (minutes to hours)
Duration of action	Brief (milliseconds)	Long (minutes to days or longer)
Major functions	Coordinates rapid, precise responses	Controls activities that require long duration rather than speed